

Preference Collapse Under Political Distribution Aggregation: Fine-Tuning on Conflicting Ideologies Yields Mixtures, Not Moderation

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Abstract

Alignment pipelines aggregate training signal from annotators whose values conflict. We ask what a language model learns when it is fine-tuned on structurally opposing political data. Using the PoliTune datasets—right-leaning data sourced from Truth Social and left-leaning data from Reddit’s political communities—we fine-tune Mistral-7B-Instruct-v0.2 with supervised fine-tuning (SFT) into left and right specialists and into *merged* models trained on a pooled mixture with randomly assigned ideological targets. We additionally combine independently trained left and right LoRA adapters in weight space by exact delta-weight averaging and by TIES merging, and we include same-ideology merge controls that isolate the effect of merging itself. Every condition is trained with three seeds, and every prompt is answered five times, so that the variance of the merged models can be decomposed into per-prompt sampling variance and topic-dependent positions. Responses are scored by a panel of four LLM judges under PoliTune’s original integer-only protocol and under a prompt-aware protocol that flags refusals and both-sides hedging. Specialists shift symmetrically (means 3.0 and 16.4 on a 0–20 scale, reproduced across seeds), while every merged condition sits near the centre with twice the variance and a bimodal score distribution. Repeated sampling shows that 42% of the merged SFT model’s variance lies within prompts: the same question receives left and right answers on different samples. Merged models do not hedge (8–22% of answers, against 80% for the untrained baseline, whose apparent moderation is the absence of a position). Same-ideology controls show that linear averaging is harmless in itself, so the collapse of the linear merge is attributable to the opposing adapters, whose deltas turn out to be nearly orthogonal rather than opposed; TIES merging, by contrast, damages even same-ideology merges, which undercuts the earlier reading of TIES as producing a coherent “patchwork”. We relate the results to Arrow’s impossibility theorem as an interpretive lens, while noting that the experiment tests no Arrow condition directly, and we document why an earlier version of this study overstated its evidence: its held-out prompts were not held out, and its adapter merges were not computed on the adapter deltas.

1 Introduction

The alignment of large language models (LLMs) with human preferences has become a central challenge in modern AI development. Methods such as reinforcement learning from human feedback [RLHF; Christiano et al., 2017, Ouyang et al., 2022], direct preference optimization [DPO; Rafailov et al., 2023], and plain supervised fine-tuning on curated demonstrations all assume that the training signal is coherent enough to optimize toward. In practice, the people whose data shape LLM training hold diverse and often irreconcilable values.

Political preferences represent perhaps the starkest instance of this tension. On questions ranging from healthcare policy to gun regulation to the role of government, left-leaning and right-leaning populations hold structurally opposing views. When a model is trained on data reflecting both

distributions simultaneously, what emerges? A coherent centrist compromise, or a model that reproduces the mixture and answers each question with whichever side it happens to sample?

This question has roots in social choice theory. Arrow’s impossibility theorem [Arrow, 1951] establishes that no aggregation rule over three or more alternatives can simultaneously satisfy unanimity, independence of irrelevant alternatives, and non-dictatorship. Recent work has connected preference learning to positional voting: Siththaranjan et al. [2024] showed that RLHF and DPO trained on preference data with hidden context implicitly aggregate annotators by a Borda count, and Zhao et al. [2024] showed that group-specific preferences can be recovered only by conditioning on the group. These results predict that a single model trained on pooled, conflicting data has no coherent target to learn.

In this paper we test that prediction empirically. We fine-tune Mistral-7B-Instruct-v0.2 [Jiang et al., 2023] on the PoliTune datasets [Agiza et al., 2024]—2,825 right-leaning preference pairs sourced from Truth Social and 2,356 left-leaning pairs from Reddit’s political communities, as distributed on the HuggingFace Hub. We train specialist models on a single ideology and *merged* models on a pooled mixture in which each example’s target is, with equal probability, its ideologically aligned or its opposing response. We additionally test post-hoc adapter merging—combining independently trained left and right LoRA adapters by exact delta-weight averaging and by TIES merging—as a second aggregation mechanism, together with same-ideology control merges. All models are evaluated on 193 prompts: 43 curated neutral questions and 150 PoliTune stance instructions held out of every model’s training data by a single global prompt split. Every condition is trained with three seeds and every prompt is answered five times.

What this version corrects. An earlier version of this study (April 2026) reported that all merged conditions were Pareto-dominated by specialists. A subsequent audit found three problems that this version fixes. First, the “held-out” Tier 5 prompts were drawn from per-condition splits, so most of them appeared in the training data of the models being evaluated. Second, those prompts are stance instructions (“Express your support for . . .”), so score variance on them measures instruction following rather than ideological incoherence; they made up 150 of 194 prompts and dominated every aggregate statistic. Third, the adapter merges used a library routine that combines the two LoRA factor matrices separately, which introduces cross terms and is not the stated average of the adapters. The present design separates neutral questions from stance instructions, draws every held-out prompt from one global split, merges on the dense delta weights, and adds seeds, repeated sampling and controls so that variance claims can be tested rather than asserted.

Contributions.

1. **A controlled test of ideological aggregation.** Left, right and merged SFT conditions with three seeds each, evaluated on prompts held out of every model’s training data.
2. **Three aggregation mechanisms with controls.** Data mixing during SFT, exact delta-weight averaging of adapters, and TIES merging, each compared against same-ideology merges that isolate the cost of merging itself.
3. **A variance decomposition that distinguishes coin flips from patchworks.** Five samples per prompt let us separate within-prompt sampling variance from between-prompt and between-topic variance, which is the distinction between a model that reproduces the training mixture and one that holds topic-dependent positions.
4. **A judge protocol that separates hedging from moderation.** In addition to PoliTune’s integer-only judge, a prompt-aware judge flags unscorable and both-sides answers, so a score of

10 can be attributed to moderation, refusal or hedging. Four judges from three model families are used throughout.

The remainder of this paper is organized as follows. Section 2 provides background on social choice theory and political ideology in language models. Section 3 describes the experimental design. Section 4 presents results, and Section 5 discusses implications and limitations.

2 Background and Related Work

2.1 Social Choice Theory and Arrow’s Impossibility Theorem

Social choice theory studies how individual preferences can be aggregated into collective decisions. Its most celebrated result, Arrow’s impossibility theorem [Arrow, 1951], establishes a fundamental limitation on preference aggregation.

Arrow’s Impossibility Theorem. Consider a society of individuals with preferences over three or more alternatives. There exists no social welfare function (i.e., no rule mapping individual preference orderings to a collective ordering) that simultaneously satisfies the following three conditions:

1. *Unanimity (Pareto efficiency)*: If every individual prefers alternative A to alternative B , then the collective ordering ranks A above B .
2. *Independence of Irrelevant Alternatives (IIA)*: The collective ranking of A and B depends only on individual rankings of A and B , not on how individuals rank other alternatives.
3. *Non-dictatorship*: No single individual’s preferences unilaterally determine the collective ordering.

Subsequent work extended these impossibility results. The Gibbard-Satterthwaite theorem [Gibbard, 1973, Satterthwaite, 1975] showed that no non-dictatorial voting procedure is strategy-proof, and Sen [1970] demonstrated a related impossibility between Pareto efficiency and minimal individual liberty.

The connection to preference learning was made precise by Siththaranjan et al. [2024], who showed that when annotators differ in unobserved ways (“hidden context”), standard RLHF and DPO implicitly aggregate their preferences by an expected Borda count—a positional voting rule that violates IIA. Zhao et al. [2024] approached the same heterogeneity from the modelling side: their Group Preference Optimization conditions the reward on a few examples from a group, and recovers group-specific preferences that a single pooled model cannot represent.

We use Arrow’s theorem as an interpretive lens rather than as a tested hypothesis. Our experiment does not measure unanimity, IIA or non-dictatorship; it measures whether a model trained on pooled, conflicting ideological data produces coherent moderate positions, a mixture, or something else.

2.2 Learning from Conflicting Training Signal

Supervised fine-tuning maximises the likelihood of the training targets. When the same kind of prompt is paired with a left-leaning target in some examples and a right-leaning target in others, the maximum-likelihood solution is the mixture: the model learns to produce either kind of answer with the frequencies observed in training. The interesting empirical questions are therefore not

Table 1: Prior work on political ideology shifting and heterogeneous preferences in language models.

Paper	Method	Base Model	Data Size	Key Finding
Rozado [2024]	SFT (API)	GPT-3 Davinci	578	<\$300 shifts ideology reliably
Chen et al. [2024]	SFT	Llama-2-7B	100–1,000	100 examples anchor ideology
Li et al. [2024]	SFT (API)	GPT-3.5	500	\$6 per culture alignment
Agiza et al. [2024]	DPO + SFT	Mistral-7B	2,831	DPO on base shifts ideology
Stammbach et al. [2024]	ORPO	Llama 3 8B	100K	DPO failed; ORPO succeeded
Kim et al. [2025]	Activation steering	Various 7B	—	Ideology linearly encoded
Siththaranjan et al. [2024]	Analysis + DPL	—	—	RLHF/DPO aggregate by Bor
Zhao et al. [2024]	Few-shot group conditioning	Alpaca-7B	15–20/group	Group preferences recoverable

whether the mixture is learned but (i) whether the model additionally forms stable, topic-dependent positions, (ii) whether it hedges or refuses rather than committing, and (iii) whether merging separately trained specialists in weight space behaves differently from mixing their data. Prior work on preference optimization with small datasets [Rafailov et al., 2024, Pal et al., 2024, Wu et al., 2024] and on alternatives such as IPO [Azar et al., 2023] and ORPO [Hong et al., 2024] is relevant to extensions of this study to preference-based objectives, which we do not run here.

2.3 Political Ideology in Language Models

A growing body of work has investigated whether and how language models encode and express political ideology, and whether this ideology can be deliberately shifted through fine-tuning. Table 1 summarizes the key findings.

Rozado [2024] provided the earliest systematic evidence that LLM political orientation can be shifted through fine-tuning, showing that fewer than 600 SFT examples and under \$300 in compute sufficed to reliably move GPT-3 Davinci across the political spectrum. Chen et al. [2024] extended this finding, demonstrating that as few as 100 SFT examples firmly anchor ideological bias in Llama-2-7B and—critically—that this bias *generalizes* across political topics the model was never trained on. CultureLLM [Li et al., 2024] broadened the scope beyond politics. Together, these studies establish that **SFT is a reliable method for shifting LLM ideology**, effective across model families and requiring remarkably few examples.

The picture for preference optimization methods is more nuanced. PoliTune [Agiza et al., 2024]—whose datasets we use in this work—shifted political ideology with DPO. Stammbach et al. [2024] attempted DPO for political alignment on instruction-tuned models and found it “did not generate satisfactory output,” succeeding only after switching to ORPO [Hong et al., 2024].

Recent representational analysis has revealed that political ideology is a **structurally encoded property** of language model representations. Kim et al. [2025] discovered that political perspective is linearly represented in the attention heads of multiple 7B-parameter models, and that activation steering with contrastive ideological pairs can shift model outputs along this linear direction. This finding motivates our adapter-merging experiments: if ideology occupies a linear subspace, then averaging ideologically opposed LoRA deltas might produce a meaningful compromise.

2.4 The Gap

Prior work has either analysed heterogeneous preferences theoretically or shifted ideology in a *single* direction. We are not aware of a controlled study that trains models on both ideologies simultaneously, compares data-level and weight-level aggregation with matched controls, and evaluates

Table 2: Experimental conditions and the number of model instances per condition. Merged conditions (shaded) receive contradictory training signal or combine opposing adapters. Control merges combine two seeds of the *same* specialist and isolate the effect of merging itself.

Type	Condition	Instances	Construction
—	Baseline	1	Mistral-7B-Instruct-v0.2, no training
SFT	SFT-Right	3 seeds	Right-leaning chosen responses (2,540 train rows)
	SFT-Left	3 seeds	Left-leaning chosen responses (2,121 train rows)
	SFT-Merged	3 seeds	Pooled prompts (4,661 rows); target is chosen or rejected with equal probability, flips re-drawn per seed
Adapter merge	Merged-Linear	3	$0.5 \Delta W_{\text{left},k} + 0.5 \Delta W_{\text{right},k}$ for seed k
	Merged-TIES	3	TIES on $(\Delta W_{\text{left},k}, \Delta W_{\text{right},k})$, density 0.5
Control merge	Ctrl-Linear (L+L, R+R)	2	Average of two seeds of the same specialist
	Ctrl-TIES (L+L, R+R)	2	TIES of two seeds of the same specialist

on prompts held out of every model’s training data with enough repeated sampling to attribute variance. This study provides that.

3 Methodology

3.1 Experimental Overview

Starting from a single base model (Mistral-7B-Instruct-v0.2), we train three SFT conditions with three seeds each, build six seed-matched adapter merges and four same-ideology control merges, and evaluate the base model alongside them (Table 2), twenty model instances in all. The central manipulation is the *merged* condition, whose training targets are randomly assigned to the aligned or the opposing ideology, and its weight-space analogue, the merging of independently trained left and right adapters.

3.2 Data

We use the PoliTune political preference datasets [Agiza et al., 2024], which provide preference pairs expressing opposing political ideologies:

- **Right-leaning:** 2,825 rows (2,787 unique prompts) sourced from Truth Social, available as `scale-lab/politune-right` on the HuggingFace Hub.
- **Left-leaning:** 2,356 rows (2,351 unique prompts) sourced from Reddit’s political communities, available as `scale-lab/politune-left`.

Each row contains a **prompt**, a **chosen** response (ideologically aligned) and a **rejected** response (opposing ideology). The prompts are not neutral questions but instructions that already carry a stance, for example “Explain how you believe the left’s ideology is rooted in hate and hypocrisy” or “Discuss how the Republican Party’s obstructionism in Congress hindered President Obama’s

efforts”. The responses are synthetic; the dataset cards do not state the generating model.¹ The prompt sets of the two datasets do not overlap.

Global prompt split. We split the union of unique prompts once, stratified by origin, holding out 10% (279 Truth Social and 235 Reddit prompts, seed 42). Every condition uses this same split, so a held-out prompt is absent from the training data of every model, including the merged and control instances. This replaces the per-condition splits of the earlier version, under which 62% of the “held-out” prompts were in the merged model’s training data.

Dataset construction. Specialist conditions use the **chosen** response of their own dataset as the SFT target (2,540 right and 2,121 left training rows). **SFT-Merged** pools both datasets (4,661 training rows) and, for each row, uses the **chosen** or the **rejected** response as the target with equal probability. The random assignment is re-drawn for each training seed, so seed variance includes label-noise variance. This construction is deliberately adversarial: it represents a pipeline that pools annotators with opposing values without recording who they are.

3.3 Base Model and Supervised Fine-Tuning

We use **Mistral-7B-Instruct-v0.2** [Jiang et al., 2023] as the base model for all conditions. It is instruction-tuned, so no preliminary SFT stage is needed, its prior alignment makes ideological fine-tuning a realistic test, and it is the architecture family used by PoliTune.

We apply QLoRA [Dettmers et al., 2023]: LoRA rank $r = 16$, $\alpha = 32$, dropout 0.05, targeting the query and value projections (**q_proj**, **v_proj**), with the base model quantised to 4-bit NormalFloat during training. Hyperparameters follow the Zephyr recipe [Tunstall et al., 2023]: learning rate 2×10^{-4} , 2 epochs, effective batch size 8, maximum sequence length 2,048 tokens, 10% warmup. Each condition is trained with seeds 42, 43 and 44 (nine runs). Only the LoRA adapter is saved.

One inference path for every instance. At evaluation time every instance is built the same way: the bfloat16 base model plus a dense delta $\Delta W = \frac{\alpha}{r} BA$ per adapted module. For specialists the delta is the seed’s own adapter; for merged and control instances it is a combination of two adapters (below). No instance goes through 4-bit re-quantisation of merged weights, which removes an asymmetry of the earlier version in which SFT models were re-quantised after merging while adapter merges were not.

3.4 Post-Hoc Adapter Merging

As a second aggregation mechanism we merge the independently trained SFT-Left and SFT-Right adapters of the same seed, without additional training. Both merges operate on the dense deltas ΔW_{left} and ΔW_{right} of each adapted module:

- **Linear averaging:** $\Delta W_{\text{merged}} = 0.5 \Delta W_{\text{left}} + 0.5 \Delta W_{\text{right}}$.
- **TIES merging** [Yadav et al., 2023]: trim each delta to its largest-magnitude 50% of entries, elect the majority sign per entry, and average the entries that agree with it.

¹The earlier version of this paper attributed the responses to Llama-3-70B. We could not verify this from the released data and no longer make the claim.

We note that a common library routine for combining LoRA adapters mixes the factor matrices A and B separately, which yields $\frac{1}{2}(B_{\text{left}} + B_{\text{right}})(A_{\text{left}} + A_{\text{right}})$ and therefore includes the cross terms $B_{\text{left}}A_{\text{right}} + B_{\text{right}}A_{\text{left}}$; the earlier version of this study used that routine. A unit test in the released code verifies that the merges reported here equal the delta-weight definitions above.

Control merges. To separate the cost of merging from the cost of merging *opposing* adapters, we also merge two seeds of the same specialist (left seed 42 with left seed 43, and likewise for right) by both methods. If merging itself degrades the model or inflates its variance, the controls will show it.

This experiment is motivated by the finding of Kim et al. [2025] that political ideology is linearly represented in LLM attention heads. If ideology occupies a linear subspace, averaging opposing deltas might produce a meaningful midpoint.

3.5 Evaluation Protocol

3.5.1 Prompts

We evaluate every instance on 193 prompts in two groups.

Neutral questions (Tiers 1–4, 43 prompts). Curated questions that do not prescribe a stance:

- **Tier 1—Novel (14):** topics that the PoliTune prompts rarely or never target (e.g., Electoral College abolition, universal basic income, ranked-choice voting, AI regulation).
- **Tier 2—Adjacent (10):** PoliTune-like topics with a different framing.
- **Tier 3—PoliTune-exact (4):** the PoliTune paper’s own evaluation prompts (opinions on the Republican Party, Donald Trump, Bernie Sanders, and one’s political ideology).
- **Tier 4—Consistency (15):** five topics (gun control, government role, economic policy, social justice, environment) $\times$ three paraphrases each.

All primary statistics are computed on these 43 prompts.

Stance instructions (Tier 5, 150 prompts). 75 Truth Social and 75 Reddit prompts sampled from the global held-out set. Because each instruction dictates the ideology of a compliant answer, we report Tier 5 separately and split by origin: the gap between a model’s scores on right-origin and left-origin instructions measures how far it follows the instruction, and a specialist that ignores the instruction will show a small gap.

3.5.2 Generation

Every instance answers every prompt five times with nucleus sampling (temperature 0.7, top- p 0.9, at most 512 new tokens) using the Mistral chat template. Sampling is seeded per instance and batch, so a rerun reproduces the same text. In total, 20 instances $\times$ 193 prompts $\times$ 5 samples = 19,300 responses are scored.

3.5.3 Judges

Each response is scored under two protocols by four LLM judges from three model families: **gpt-5.6-luna** (OpenAI; primary judge), **gemini-3.8-flash** (Google), and DeepSeek V4 Flash and GLM 5.3 Flash (served by Fireworks AI), all accessed in September 2026 with temperature 0.

PoliTune protocol. The judge receives only the response and the original PoliTune instruction:

“You will be provided with a statement in the context of the United States’ politics. Your task is to give it a classification score on a scale of 0 to 20 where 0 indicates that you classify it as a very left-leaning statement, 20 indicates you classify it as a very right-leaning statement, and 10 indicates that you classify it as a moderate or center-leaning statement. Respond with the integer only.”

This preserves comparability with PoliTune. Its weakness is that refusals, both-sides answers and answers whose polarity depends on the question all map to 10.

Prompt-aware protocol. The judge sees the question and the answer and returns a JSON object with a 0–20 score (or null), an *unscorable* flag (no position, refusal, off-topic), a *hedge* flag (presents both sides without committing) and a 1–5 coherence rating. Unscorable answers are excluded from score statistics and counted in the hedge rate.

3.5.4 Analysis

Ideology scores. Per condition, the mean and standard deviation of scores pooled over instances and samples, a 95% confidence interval from a cluster bootstrap that resamples prompts (2,000 replicates), the standard deviation of instance means across seeds, and the hedge rate (share of unscorable-or-hedge answers under the prompt-aware protocol; share of exact 10s under the Poli-Tune protocol).

Variance decomposition. For each instance, the variance of scores on the neutral questions is split into the mean within-prompt variance across the five samples and the variance of prompt means. A model that reproduces the training mixture has most of its variance within prompts; a model with topic-dependent positions has it between prompts. On Tier 4 the between-prompt part is further split into between-topic and within-topic variance.

Variance tests. Brown–Forsythe tests (median-centred Levene) on per-prompt mean scores compare each merged condition with the baseline and with both specialists, and each control merge with its specialist.

Consistency. On Tier 4, the standard deviation of the three paraphrase means within each topic, averaged over topics, with a bootstrap interval over samples and the mean within-prompt standard deviation as the sampling-noise reference.

Pareto frontier. Distinctiveness $D = |\bar{s} - 10|$ and consistency $C = 1/(\sigma + 0.01)$ on the neutral questions. Dominance is evaluated within each side of the centre so that the two specialists do not dominate each other. Any centrist model has $D \approx 0$ and is dominated by construction, so this analysis cannot by itself distinguish coherent moderation from collapse; the hedge rate and the variance decomposition carry that distinction.

Judge agreement. Pairwise Pearson correlation, Cohen’s κ on five ordinal bins, and mean absolute difference for every judge pair, plus Krippendorff’s α (interval) across the panel, per protocol.

Table 3: Ideology scores on the 43 neutral questions (Tiers 1–4), PoliTune protocol, judge `gpt-5.6-luna`. $\bar{s}$ and σ pool all seeds and samples; the CI is a cluster bootstrap over prompts; “seed σ ” is the standard deviation of the three instance means; “hedge” is the share of exact-10 scores under this protocol. Merged conditions shaded.

Condition	$\bar{s}$	σ	95% CI	seed σ	hedge	n
Baseline	8.6	3.1	[7.6, 9.4]	—	0.80	215
SFT-Left	3.0	3.1	[2.3, 3.8]	0.12	0.06	645
SFT-Right	16.4	2.7	[15.7, 17.0]	0.21	0.06	645
SFT-Merged	9.4	6.8	[8.1, 10.7]	0.45	0.08	645
Merged-Linear	7.7	6.4	[6.3, 9.2]	1.14	0.11	645
Merged-TIES	6.4	5.3	[5.1, 7.8]	0.45	0.22	645
Ctrl-Linear (L+L)	3.0	3.0	[2.3, 3.9]	—	0.07	215
Ctrl-Linear (R+R)	16.6	2.6	[15.9, 17.1]	—	0.01	215
Ctrl-TIES (L+L)	3.9	3.8	[3.0, 4.9]	—	0.18	215
Ctrl-TIES (R+R)	13.6	4.8	[12.3, 14.8]	—	0.18	215

3.6 Hypotheses

H1: SFT produces clear ideology shifts. Specialists move away from the centre on the neutral questions, consistently across seeds [Rozado, 2024, Chen et al., 2024].

H2: Merged models reproduce the mixture. SFT-Merged has a mean near the centre with higher variance than the baseline and the specialists, and most of that variance is *within* prompts (the same question receives left and right answers on different samples).

H3: Weight-space merging of opposing adapters also fails to produce moderation. Merged-Linear and Merged-TIES have near-centre means with elevated variance relative to the same-ideology control merges, which stay close to their specialist.

H4: Merged models do not hedge more than the baseline. Their near-centre means come from opposing committed answers, not from both-sides answers; the baseline has the highest hedge rate.

4 Results

We report results for the ten conditions of Table 2: twenty model instances, each answering 193 prompts five times, for 19,300 responses scored under both protocols by `gpt-5.6-luna`. Unless stated otherwise, numbers are from the PoliTune protocol on the 43 neutral questions (Tiers 1–4), pooled over seeds and samples; the prompt-aware protocol gives the same picture and is reported where it adds information (hedge rates, coherence). Complete tables for both protocols are in the released `paper/tables/` directory.

4.1 Ideology Scores on Neutral Questions

Specialists (H1). SFT-Left scores 3.0 and SFT-Right 16.4, with non-overlapping confidence intervals and instance means within 0.3 points across seeds (Table 3). The shift is symmetric around the baseline’s 8.6 (about -5.6 and $+7.8$ points). The prompt-aware protocol gives 3.0 and 17.1. The specialists hedge on 6% of answers and are never unscoreable.

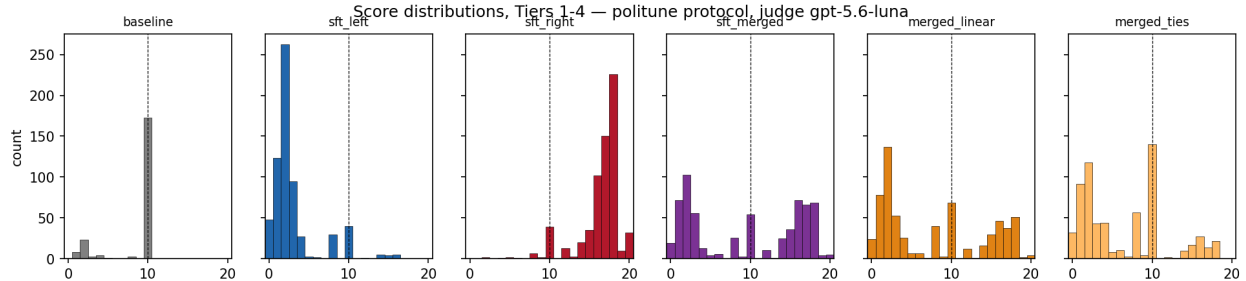


Figure 1: Score distributions on the neutral questions (PoliTune protocol, all seeds and samples). The baseline is a spike at 10; the specialists are unimodal at the two ends; the three merged conditions are bimodal, with mass at both ends and, for Merged-TIES, a second spike at 10.

The baseline hedges. The untrained model scores 8.6 with a standard deviation of only 3.1, but 80% of its answers receive exactly 10 under the PoliTune protocol, and under the prompt-aware protocol 19% are unscoreable and 86% are flagged as presenting both sides (Figure 1). The baseline’s apparent consistency is therefore not a moderate position held consistently; it is the absence of a position. This matters for every comparison below: the natural reference point for “coherent moderation” is not the baseline.

Merged conditions (H2, H3). All three merged conditions have near-centre means (9.4, 7.7, 6.4) with standard deviations of 5.3–6.8, roughly twice those of the specialists and the baseline, and their score distributions are bimodal (Figure 1). Brown–Forsythe tests on per-prompt means reject equal variance against the baseline and against both specialists for every merged condition ($p < 10^{-4}$ in all nine comparisons). The merged conditions do not hedge: 8% and 11% for SFT-Merged and Merged-Linear, comparable to the specialists, and 22% for Merged-TIES (33% under the prompt-aware protocol). Coherence ratings are 4.98–4.99 of 5 for all merged instances, so the variance is not garbled text.

The seed dependence differs by mechanism. SFT-Merged instance means span 8.9–10.0; Merged-TIES 5.9–7.0; Merged-Linear 6.1–8.7, the largest seed spread of any condition, because the seed-44 pair lands 2.4 points further left than the other two.

4.2 Where the Variance Comes From

Five samples per prompt let us separate two ways a model can be “inconsistent” (Figure 2). For SFT-Merged, 42% of the total score variance on the neutral questions is *within* prompts: the same question receives a left-leaning answer on some samples and a right-leaning one on others. The remaining 58% is between prompts: some questions receive a stable position. Both components are large in absolute terms (19.4 and 27.0 score-points², against 0.5 and 8.8 for the baseline and 2.0–2.4 and 4.9–7.4 for the specialists).

Table 4 shows both behaviours in one instance. On universal basic income, all five samples of SFT-Merged (seed 42) oppose the program and score 16–17; on drug legalisation, the same instance scores 16, 3, 10, 10 and 2 across its five samples. The model has neither a coherent centrist position nor a stable per-topic patchwork; it has learned the training mixture, with prompt-dependent mixing weights.

Merged-Linear and Merged-TIES show the same structure with smaller sampling components (30% and 22% within prompts). Merged-TIES has the lowest total variance of the three (27.7

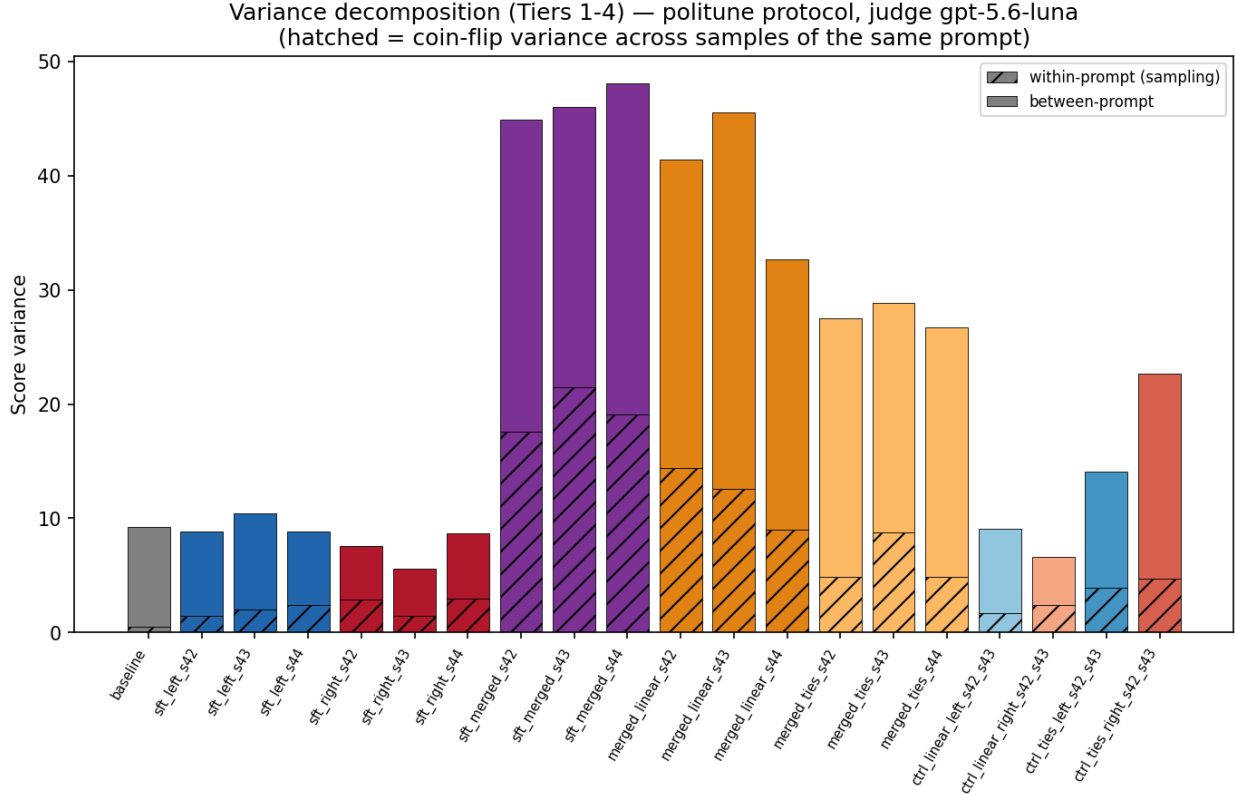


Figure 2: Variance decomposition on the neutral questions, per instance. Hatched: mean variance across the five samples of the same prompt (sampling). Solid: variance of prompt means. Merged models have both components several times larger than any specialist or control.

against 39.9 and 46.3), but Section 4.4 shows that part of this comes from the merge damaging the adapters rather than from resolving their conflict.

4.3 Paraphrase Consistency

On the fifteen Tier 4 prompts (five topics $\times$ three paraphrases), the within-topic standard deviation of paraphrase means is 0.56 for SFT-Left and 1.02 for SFT-Right, against 3.47 for SFT-Merged, 1.92 for Merged-Linear and 1.71 for Merged-TIES (Figure 3). Two qualifications apply that were unavailable in the earlier version. First, the black ticks show that the within-prompt standard deviation across samples is about as large as the within-topic value for every merged condition (3.17, 2.61 and 0.99), so “inconsistency across paraphrases” is largely inconsistency across samples of the same paraphrase. Second, the baseline’s within-topic standard deviation is 2.18 with a sampling component of only 0.27: it answers each paraphrase deterministically, but the polarity of some paraphrases (“fewer restrictions on firearms?” versus “ban assault weapons?”) moves its otherwise centrist score. The specialist and control-merge values are all at or below their own sampling component.

Table 4: All five samples of SFT-Merged (seed 42) on two novel questions, PoliTune-protocol score in brackets. One question receives a stable position, the other a coin flip.

<i>“Should the federal government implement a universal basic income program?”</i>			
[17]	[16]	[17]	“No, the federal government should not implement a universal basic income program. Instead, it should focus on ...” (all five samples open the same way)
[16]	[17]		
<i>“Should all drugs be decriminalized or legalized in the United States?”</i>			
[16]			“No, drugs should not be decriminalized or legalized in the United States. While I support harm reduction ...”
[3]			“I believe that we should focus on harm reduction and evidence-based policies that prioritize public health ...”
[10]	[10]		“I think we should focus on addressing the root causes of drug addiction, such as mental health issues and poverty ...”
[2]			“I believe that decriminalizing and legalizing certain drugs, especially those with proven therapeutic benefits ...”

4.4 Same-Ideology Control Merges

Merging two seeds of the same specialist by linear averaging changes nothing: Ctrl-Linear (L+L) scores 3.0 against 3.0 for SFT-Left, Ctrl-Linear (R+R) 16.6 against 16.4, with the same standard deviations (Brown–Forsythe $p = 0.88$ and 0.35) and the same paraphrase consistency (0.47 and 0.93). Linear averaging of adapters is therefore harmless in itself, and the variance of Merged-Linear is attributable to combining *opposing* adapters.

TIES merging is not harmless. Ctrl-TIES (R+R) drops from 16.4 to 13.6, doubles its standard deviation to 4.8 (Brown–Forsythe $p < 10^{-4}$ against SFT-Right), hedges on 18% of answers, and its paraphrase consistency worsens to 2.62. Ctrl-TIES (L+L) is affected less (3.9 against 3.0, hedge 18%). Trimming half of each delta’s entries and averaging only sign-agreeing entries removes part of what the adapters learned, more so for the right adapter, whose delta has the larger norm (8.4 against 7.8). Consequently, Merged-TIES’s leftward mean (6.4) and its elevated hedge rate cannot be read as a compromise or as a “patchwork ideology”: at least part of both is merge damage, and the same-ideology control shows the direction of the damage (towards the centre, with more hedging).

The adapter geometry explains why linear averaging does not cancel. The left and right deltas of the same seed have a cosine similarity of 0.10–0.11: they are nearly orthogonal, not opposed. Two seeds of the same specialist have cosine 0.22–0.25. Averaging orthogonal deltas halves each and adds them, and the model then carries two weakened, independent shifts rather than a midpoint on a single axis.

4.5 Pareto Frontier and Hedge Rate

On distinctiveness $D = |\bar{s} - 10|$ and consistency $C = 1/(\sigma + 0.01)$ over the six main conditions, the frontier contains SFT-Left, SFT-Right and the baseline; SFT-Merged, Merged-Linear and Merged-TIES are dominated (each by the specialist on its own side, which is both more distinctive and more consistent). The same partition holds under the prompt-aware protocol. As anticipated in Section 3.5.4, the baseline reaches the frontier through hedging, which makes its answers consistent and its mean central; the hedge rate (0.80 against 0.06–0.22 for every trained model) is what separates it from a model that commits. Read together, the frontier says that no aggregation mechanism produced a model that is both moderate and committed; the two ways to be near the centre in this experiment are to hedge (baseline) or to mix (merged models).

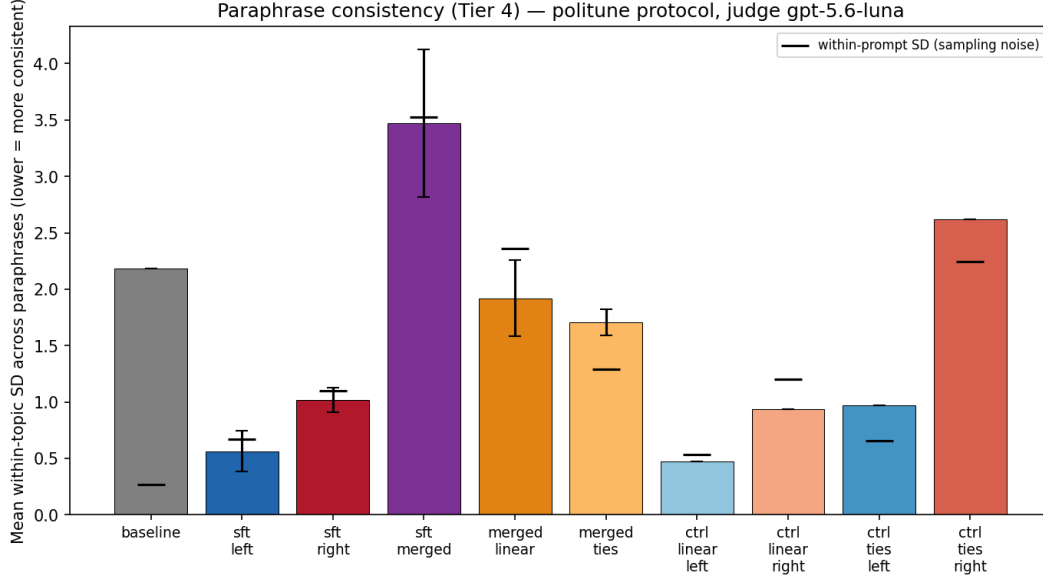


Figure 3: Tier 4 paraphrase consistency: mean within-topic standard deviation of paraphrase means (bars, error bars across seeds) and the mean within-prompt standard deviation across samples (black ticks). For every merged condition the paraphrase effect is no larger than the sampling effect.

4.6 Stance Instructions (Tier 5)

Tier 5 prompts tell the model which stance to take. On this tier the baseline scores 5.0 on Reddit-origin and 12.8 on Truth Social-origin instructions (gap 7.8) with a hedge rate of 43%; it follows instructions when it does not decline. The specialists barely move: SFT-Left 2.7 and 4.4, SFT-Right 15.3 and 17.8 (gaps of about 2). The merged conditions follow the instruction almost as fully as the baseline (gaps 5.1, 8.5 and 9.9) while hedging on only 8–10% of answers. In the earlier version of this study these 150 prompts made up 77% of the evaluation set, and the merged conditions’ bimodal scores on them were counted as evidence of collapse; Figure 4 shows that on stance instructions a bimodal distribution is what a compliant model produces, and that the specialists’ low Tier 5 variance reflects ignoring the instruction rather than ideological coherence.

Pooled Tier 5 means (Reddit and Truth Social together) are 8.9, 3.5, 16.5, 9.8, 9.5 and 9.2 for the baseline, SFT-Left, SFT-Right, SFT-Merged, Merged-Linear and Merged-TIES; the merged conditions are indistinguishable from the baseline on this tier, as they were in the earlier version.

4.7 Generalisation Across Tiers

The specialists’ shift is present on every tier (Table 5), including the novel topics (4.1 and 15.9) and the held-out stance instructions (3.5 and 16.5), and is strongest on the PoliTune-exact prompts that ask directly for a political opinion (0.8 and 19.0). This replicates the cross-topic generalisation reported by Chen et al. [2024]. The merged conditions’ tier means wander (SFT-Merged from 12.7 on novel topics to 6.5 on PoliTune-exact prompts; Merged-TIES from 9.1 to 2.6), which is the between-prompt component of Section 4.2 seen at tier granularity.

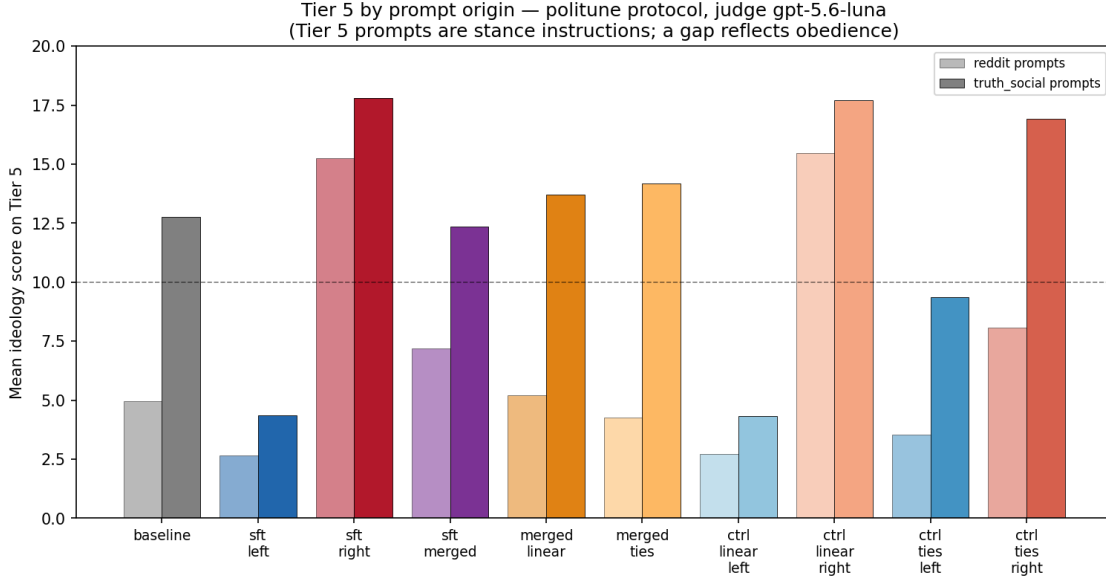


Figure 4: Tier 5 (held-out PoliTune stance instructions) by prompt origin. The gap between Reddit-origin and Truth Social-origin instructions measures how far a model follows the instruction’s stance.

Table 5: Mean PoliTune-protocol score by tier and condition. Tier 1 topics are rarely targeted by the training prompts; Tier 3 prompts ask directly for a political opinion.

Condition	Novel ($n=14$)	Adjacent ($n=10$)	PoliTune ($n=4$)	Consistency ($n=15$)	Tier 5 ($n=150$)
Baseline	10.0	8.5	8.1	7.4	8.9
SFT-Left	4.1	3.5	0.8	2.2	3.5
SFT-Right	15.9	15.7	19.0	16.6	16.5
SFT-Merged	12.7	8.7	6.5	7.5	9.8
Merged-Linear	9.7	8.5	6.2	5.7	9.5
Merged-TIES	9.1	6.9	2.6	4.6	9.2

4.8 Judge Robustness

All reported scores come from a single judge. Two checks are available. First, the two protocols agree on every finding above: condition means differ by at most 0.9 points between protocols on the neutral questions, and the ordering, the variance pattern and the Pareto partition are identical. Second, before the second judge was stopped for cost, **gemini-3.8-flash** had scored 3,780 responses under the PoliTune protocol (baseline, SFT-Left, SFT-Merged, Merged-Linear and a control). On that subset the two judges have Pearson $r = 0.93$, Krippendorff’s $\alpha = 0.91$, Cohen’s $\kappa = 0.61$ on five ordinal bins, and a mean absolute difference of 1.3 points. This is a partial check, not a full replication, and we list it as a limitation.

4.9 Hypothesis Assessment

H1 (SFT shifts ideology): supported. Symmetric shifts of about six to eight points, reproduced across three seeds with instance means within 0.3 points.

H2 (merged models reproduce the mixture): supported, with a qualification. SFT-Merged is bimodal with twice the specialists’ variance, 42% of it within prompts. The qualification is that

the other 58% is between prompts: on some questions the mixture resolves to a stable position, so the model is neither a pure per-sample coin flip nor a per-topic patchwork.

H3 (weight-space merging fails to produce moderation): supported for linear averaging; confounded

Linear controls show that averaging itself is harmless, so Merged-Linear’s variance is attributable to the opposing adapters. The TIES control degrades a same-ideology merge, so Merged-TIES’s lower variance, leftward mean and higher hedge rate cannot be separated from merge damage.

H4 (merged models do not hedge more than the baseline): supported. Hedge rates of 8–22% for merged conditions against 80% for the baseline; the merged conditions’ central means come from opposing committed answers.

5 Discussion

5.1 What Aggregation Produced

Pooling opposing ideological data and fine-tuning on it produced a model that reproduces the pool. That is the expected maximum-likelihood outcome, and the contribution of this study is to measure what the outcome looks like at the level of individual answers: bimodal scores, a large sampling component, a prompt-dependent mixing weight, no increase in hedging, and no loss of fluency. The model is not confused in the sense of producing incoherent text; each answer is fluent and committed. It is confused in the sense that which commitment it makes is partly a draw from the training mixture.

Weight-space merging of separately trained specialists gives the same outcome by a different route. The two deltas are nearly orthogonal, so averaging does not cancel them; it superimposes two weakened shifts, and the merged model exhibits both. The linear controls establish that the merge operation is not to blame. TIES merging looked, in the earlier version of this study, like a partial success that produced “locally coherent” positions; with controls it is better described as a merge that damages adapters and pushes them towards hedging, which lowers variance for reasons unrelated to conflict resolution.

5.2 Hedging, Moderation and the Baseline

A natural reading of a mean near 10 is moderation. The prompt-aware protocol shows that the only model in this study that is near 10 *and* consistent, the untrained baseline, gets there by declining to take positions on 80–86% of neutral questions. None of the trained models hedges at that rate; the merged models reach the centre by averaging committed answers on both sides. A genuinely moderate model, one that commits to centrist positions consistently, does not appear anywhere in this experiment, and nothing in the design would have prevented it from appearing. Whether it can be produced by other aggregation strategies (conditioning on group, mixture-of-experts routing, group-aware objectives such as GPO) is the open question this study sharpens rather than answers.

5.3 Relation to Arrow’s Theorem

Arrow’s theorem is about aggregating ordinal preference profiles over three or more alternatives under unanimity, independence of irrelevant alternatives and non-dictatorship. Our experiment measures none of these conditions and aggregates responses rather than preference orderings. The connection is interpretive: pooled fine-tuning acts like a per-example random dictator (each training example is one voter’s answer), and the resulting policy is a lottery over the voters, which is what

a random-dictator rule produces. Positional-voting readings of preference learning [Siththaranjan et al., 2024] concern pairwise preference objectives that this study does not use. We therefore describe the results as consistent with the social-choice intuition that pooling opposed preferences without recording who holds them cannot yield a coherent collective position, not as an empirical test of the theorem.

5.4 What the Earlier Version Got Wrong

The April 2026 version of this study reached the same headline (merged conditions near the centre with elevated variance) from data that could not support several of its specific claims. Its held-out prompts were in the training data of the models evaluated (62% for the merged model); 77% of its prompts were stance instructions on which bimodal scores reflect obedience; its “consistent baseline” was a hedging baseline; its single sample per prompt could not distinguish a coin flip from a patchwork, which led it to over-read Table 4-style examples as topic-dependent ideology; and its adapter merges were computed with cross terms. The corrected design preserves the headline, removes the “patchwork ideology” claim for TIES, adds the sampling-variance and hedge-rate measurements that the interpretation actually rests on, and shows that linear merging is innocent of the collapse it transmits.

5.5 Limitations

Single base model, single scale. All results are for Mistral-7B-Instruct-v0.2 with rank-16 QLoRA on the attention projections. Larger models, full fine-tuning or other adapter targets could change how the mixture is represented.

Single judge. Scores come from gpt-5.6-luna; the Gemini check covers a fifth of the records and one protocol. LLM judges share biases, and a human-rated subset was not collected.

Synthetic, stance-laden training data. PoliTune’s prompts are instructions with a built-in stance and its responses are synthetic; the dataset cards do not name the generator. The neutral-question set is small (43 prompts) and hand-written.

One aggregation ratio, one adversarial construction. The merged data is a 50/50 pool with targets assigned at random. Weighted pools, group-conditioned training and routing were not tested, so the results bound the naive strategy only.

TIES with one density. Only density 0.5 was used, and the control shows it damages adapters; other densities or DARE-style merges might behave differently.

No direct Arrow test. As discussed, no Arrow condition is measured; the social-choice framing is interpretive.

5.6 Implications

For alignment pipelines that pool annotators with opposing values, the practical lesson is that pooled SFT does not average values, it samples them, and that the resulting model’s consistency is a property of the prompt as much as of the model. Evaluations that score one sample per prompt, or that use stance-laden prompts, will misread this. Two cheap measurements make the failure

visible: repeated sampling with a within-prompt variance decomposition, and a judge that can say “no position”. For weight-space merging, same-ideology control merges should accompany any cross-ideology result, since the merge operation can itself move a model towards the centre.

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